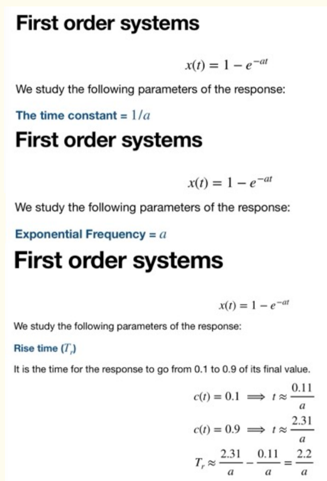


First order systems

$$x(t) = 1 - e^{-at}$$

We study the following parameters of the response:

The time constant = $1/a$

First order systems

$$x(t) = 1 - e^{-at}$$

We study the following parameters of the response:

Exponential Frequency = a

First order systems

$$x(t) = 1 - e^{-at}$$

We study the following parameters of the response:

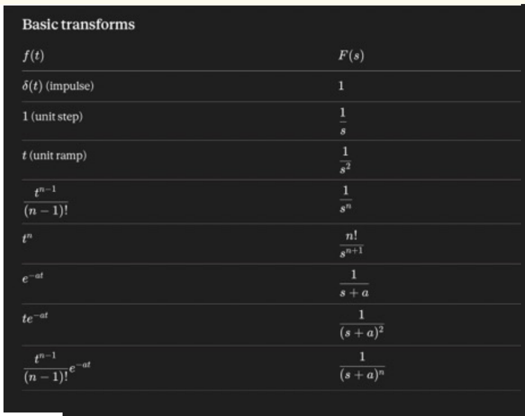
Rise time (T_r)

It is the time for the response to go from 0.1 to 0.9 of its final value.

$$c(t) = 0.1 \implies t \approx \frac{0.11}{a}$$

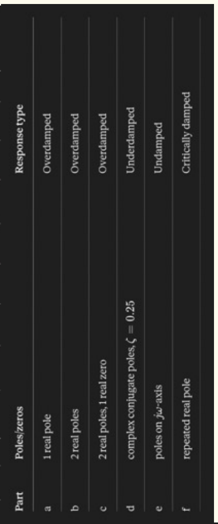
$$c(t) = 0.9 \implies t \approx \frac{2.31}{a}$$

$$T_r \approx \frac{2.31}{a} - \frac{0.11}{a} = \frac{2.2}{a}$$

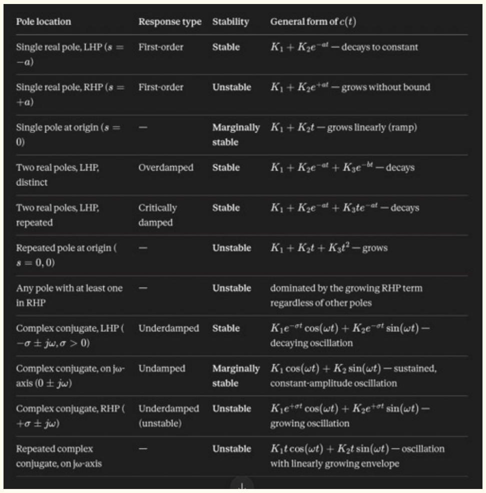


Basic transforms

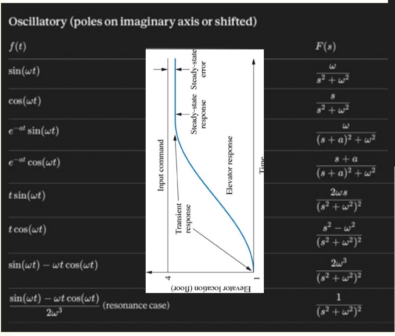
$f(t)$	$F(s)$
$\delta(t)$ (impulse)	1
1 (unit step)	$\frac{1}{s}$
t (unit ramp)	$\frac{1}{s^2}$
$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$
t^n	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{t^{n-1}}{(n-1)!}e^{-at}$	$\frac{1}{(s+a)^n}$



Part	Poles/zeros	Response type
a	1 real pole	Overdamped
b	2 real poles	Overdamped
c	2 real poles, 1 real zero	Overdamped
d	complex conjugate poles, $\zeta = 0.25$	Underdamped
e	poles on $j\omega$-axis	Undamped
f	repeated real pole	Critically damped



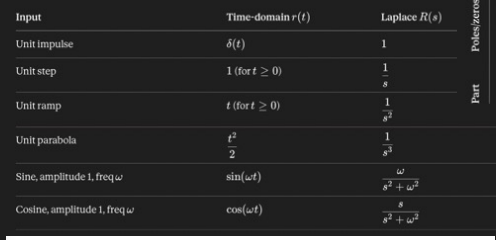
Pole location	Response type	Stability	General form of $c(t)$
Single real pole, LHP ($s = -a$)	First-order	**Stable**	$K_1 + K_2e^{-at}$ — decays to constant
Single real pole, RHP ($s = +a$)	First-order	**Unstable**	$K_1 + K_2e^{+at}$ — grows without bound
Single pole at origin ($s = 0$)	—	**Marginally stable**	$K_1 + K_2t$ — grows linearly (ramp)
Two real poles, LHP, distinct	Overdamped	**Stable**	$K_1 + K_2e^{-at} + K_3e^{-bt}$ — decays
Two real poles, LHP, repeated	Critically damped	**Stable**	$K_1 + K_2e^{-at} + K_3te^{-at}$ — decays
Repeated pole at origin ($s = 0, 0$)	—	**Unstable**	$K_1 + K_2t + K_3t^2$ — grows
Any pole with at least one in RHP	—	**Unstable**	dominated by the growing RHP term regardless of other poles
Complex conjugate, LHP ($-\sigma \pm j\omega, \sigma > 0$)	Underdamped	**Stable**	$K_1e^{-\sigma t}\cos(\omega t) + K_2e^{-\sigma t}\sin(\omega t)$ — decaying oscillation
Complex conjugate, on $j\omega$-axis ($0 \pm j\omega$)	Undamped	**Marginally stable**	$K_1\cos(\omega t) + K_2\sin(\omega t)$ — sustained, constant-amplitude oscillation
Complex conjugate, RHP ($+\sigma \pm j\omega$)	Underdamped (unstable)	**Unstable**	$K_1e^{+\sigma t}\cos(\omega t) + K_2e^{+\sigma t}\sin(\omega t)$ — growing oscillation
Repeated complex conjugate, on $j\omega$-axis	—	**Unstable**	$K_1t\cos(\omega t) + K_2t\sin(\omega t)$ — oscillation with linearly growing envelope



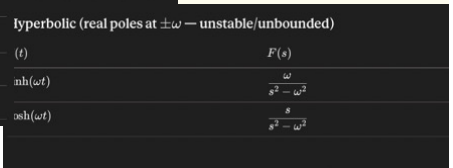
Oscillatory (poles on imaginary axis or shifted)

$f(t)$	$F(s)$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$
$t\sin(\omega t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t\cos(\omega t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$\sin(\omega t) - \omega t\cos(\omega t)$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$
$\frac{\sin(\omega t) - \omega t\cos(\omega t)}{2\omega^3}$ (resonance case)	$\frac{1}{(s^2+\omega^2)^2}$

Steady-state error
Steady-state response
Input command
Transient response
Elevator response
Time
Elevator location (floor)
4
1

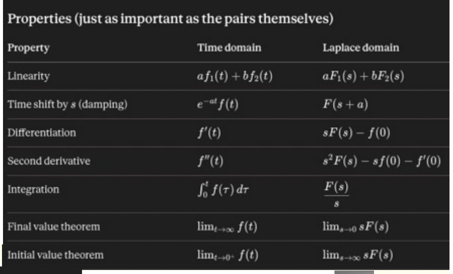


Input	Time-domain $r(t)$	Laplace $R(s)$
Unit impulse	$\delta(t)$	1
Unit step	1 (for $t \ge 0$)	$\frac{1}{s}$
Unit ramp	t (for $t \ge 0$)	$\frac{1}{s^2}$
Unit parabola	$\frac{t^2}{2}$	$\frac{1}{s^3}$
Sine, amplitude 1, freq ω	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
Cosine, amplitude 1, freq ω	$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$



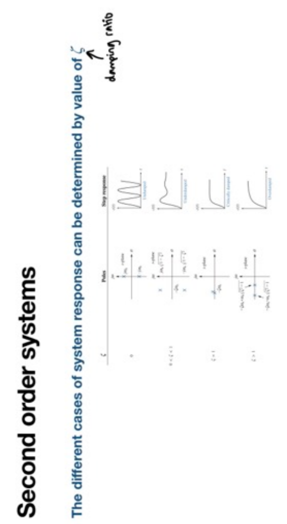
yperbolic (real poles at $\pm\omega$ — unstable/unbounded)

(t)	$F(s)$
$\text{inh}(\omega t)$	$\frac{\omega}{s^2-\omega^2}$
$\text{osh}(\omega t)$	$\frac{s}{s^2-\omega^2}$



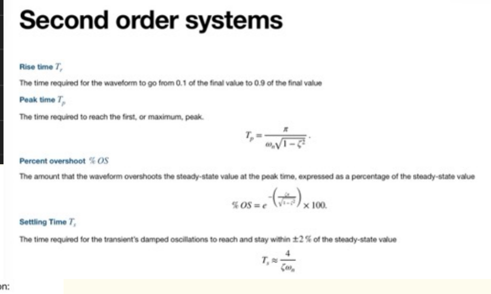
Properties (just as important as the pairs themselves)

Property	Time domain	Laplace domain
Linearity	$af_1(t) + bf_2(t)$	$aF_1(s) + bF_2(s)$
Time shift by s (damping)	$e^{-at}f(t)$	$F(s+a)$
Differentiation	$f'(t)$	$sF(s) - f(0)$
Second derivative	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$
Integration	$\int_0^t f(\tau)\,d\tau$	$\frac{F(s)}{s}$
Final value theorem	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$
Initial value theorem	$\lim_{t\to 0^+} f(t)$	$\lim_{s\to\infty} sF(s)$



Second order systems

The different cases of system response can be determined by value of ζ ← damping ratio



Second order systems

Rise time T_r

The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value

Peak time T_p

The time required to reach the first, or maximum, peak.

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}.$$

Percent overshoot $\% OS$

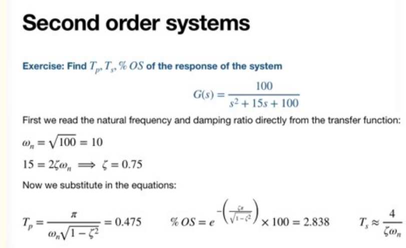
The amount that the waveform overshoots the steady-state value at the peak time, expressed as a percentage of the steady-state value

$$\% OS = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100.$$

Settling Time T_s

The time required for the transient's damped oscillations to reach and stay within ±2 % of the steady-state value

$$T_s \approx \frac{4}{\zeta\omega_n}$$



Second order systems

Exercise: Find $T_p, T_s, \% OS$ of the response of the system

$$G(s) = \frac{100}{s^2 + 15s + 100}$$

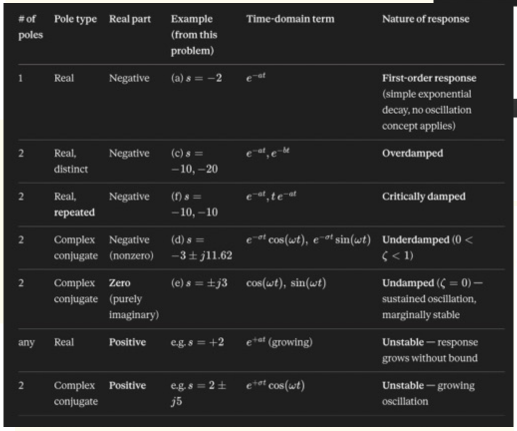
First we read the natural frequency and damping ratio directly from the transfer function:

$$\omega_n = \sqrt{100} = 10$$

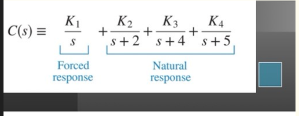
$$15 = 2\zeta\omega_n \implies \zeta = 0.75$$

Now we substitute in the equations:

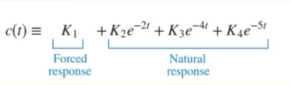
$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 0.475 \qquad \% OS = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100 = 2.838 \qquad T_s \approx \frac{4}{\zeta\omega_n} = 0.533$$



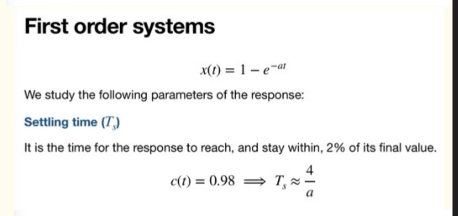
# of poles	Pole type	Real part	Example (from this problem)	Time-domain term	Nature of response
1	Real	Negative	(a) $s = -2$	e^{-at}	**First-order response** (simple exponential decay, no oscillation concept applies)
2	Real, distinct	Negative	(c) $s = -10, -20$	e^{-at}, e^{-bt}	**Overdamped**
2	Real, **repeated**	Negative	(f) $s = -10, -10$	$e^{-at}, t\,e^{-at}$	**Critically damped**
2	Complex conjugate	Negative (nonzero)	(d) $s = -3 \pm j11.62$	$e^{-\sigma t}\cos(\omega t),\ e^{-\sigma t}\sin(\omega t)$	**Underdamped** ($0 < \zeta < 1$)
2	Complex conjugate	**Zero** (purely imaginary)	(e) $s = \pm j3$	$\cos(\omega t),\ \sin(\omega t)$	**Undamped** ($\zeta = 0$) — sustained oscillation, marginally stable
any	Real	**Positive**	e.g. $s = +2$	e^{+at} (growing)	**Unstable** — response grows without bound
2	Complex conjugate	**Positive**	e.g. $s = 2 \pm j5$	$e^{+\sigma t}\cos(\omega t)$	**Unstable** — growing oscillation



$$C(s) \equiv \underbrace{\frac{K_1}{s}}_{\text{Forced response}} + \underbrace{\frac{K_2}{s+2} + \frac{K_3}{s+4} + \frac{K_4}{s+5}}_{\text{Natural response}}$$



$$c(t) \equiv \underbrace{K_1}_{\text{Forced response}} \underbrace{+ K_2e^{-2t} + K_3e^{-4t} + K_4e^{-5t}}_{\text{Natural response}}$$



First order systems

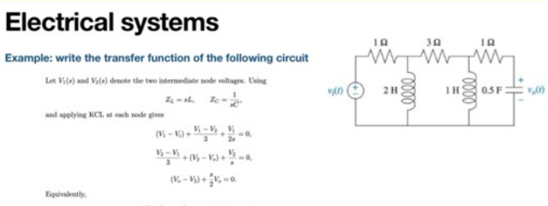
$$x(t) = 1 - e^{-at}$$

We study the following parameters of the response:

Settling time (T_s)

It is the time for the response to reach, and stay within, 2% of its final value.

$$c(t) = 0.98 \implies T_s \approx \frac{4}{a}$$



Electrical systems

Example: write the transfer function of the following circuit

Let $V_1(s)$ and $V_2(s)$ denote the two intermediate node voltages. Using

$$Z_L = sL, \qquad Z_C = \frac{1}{sC},$$

and applying KCL at each node gives

$$(V_1 - V_i) + \frac{V_1 - V_2}{3} + \frac{V_1}{2s} = 0,$$

$$\frac{V_2 - V_1}{3} + (V_2 - V_o) + \frac{V_2}{s} = 0,$$

$$(V_o - V_2) + \frac{s}{2}V_o = 0.$$

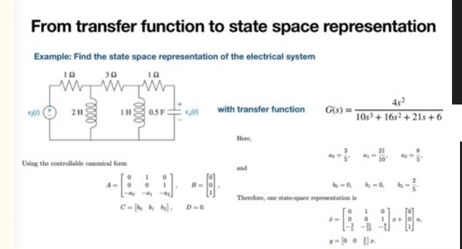
Equivalently,

$$\begin{bmatrix} \frac{4}{3} + \frac{1}{2s} & -\frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{4}{3} + \frac{1}{s} & -1 \\ 0 & -1 & 1 + \frac{s}{2} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_o \end{bmatrix} = \begin{bmatrix} V_i \\ 0 \\ 0 \end{bmatrix}.$$

Solving for $V_o(s)$ yields

$$\frac{V_o(s)}{V_i(s)} = \frac{4s^2}{10s^3 + 16s^2 + 21s + 6}$$

1 Ω
3 Ω
1 Ω
$v_i(t)$
2 H
1 H
0.5 F
$v_o(t)$



From transfer function to state space representation

Example: Find the state space representation of the electrical system

with transfer function $$G(s) = \frac{4s^2}{10s^3 + 16s^2 + 21s + 6}$$

1 Ω
3 Ω
1 Ω
$v_i(t)$
2 H
1 H
0.5 F
$v_o(t)$

Using the controllable canonical form

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$C = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix}, \qquad D = 0.$$

Here,

$$a_0 = \frac{3}{5}, \quad a_1 = \frac{21}{10}, \quad a_2 = \frac{8}{5},$$

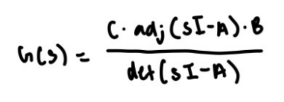
and

$$b_0 = 0, \quad b_1 = 0, \quad b_2 = \frac{2}{5}.$$

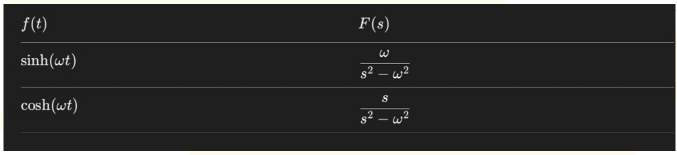
Therefore, one state-space representation is

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{3}{5} & -\frac{21}{10} & -\frac{8}{5} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u,$$

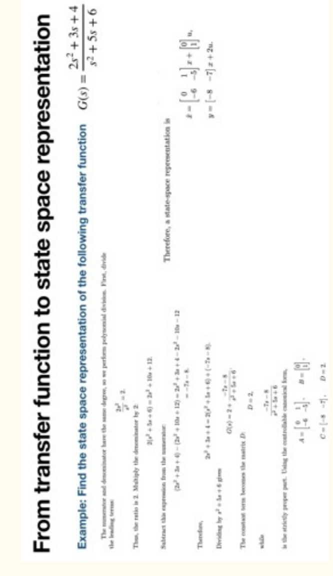
$$y = \begin{bmatrix} 0 & 0 & \frac{2}{5} \end{bmatrix} x.$$



$$G(s) = \frac{C \cdot \text{adj}(sI - A) \cdot B}{\det(sI - A)}$$

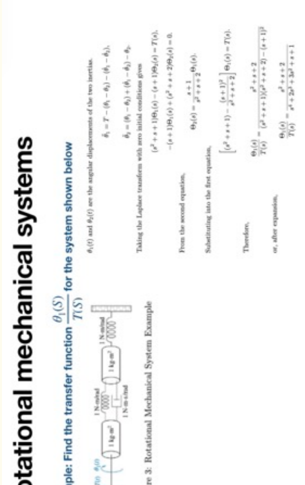


$f(t)$	$F(s)$
$\sinh(\omega t)$	$\frac{\omega}{s^2-\omega^2}$
$\cosh(\omega t)$	$\frac{s}{s^2-\omega^2}$



From transfer function to state space representation

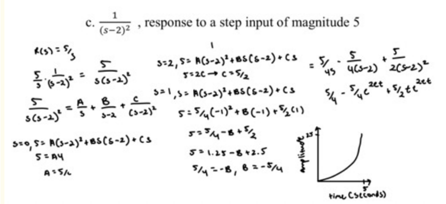
Example: Find the state space representation of the following transfer function $G(s) = \frac{2s^2 + 3s + 4}{s^2 + 5s + 6}$



Rotational mechanical systems

Example: Find the transfer function $\frac{\theta_2(S)}{T(S)}$ for the system shown below

Figure 3: Rotational Mechanical System Example



c. $\frac{1}{(s-2)^2}$, response to a step input of magnitude 5

$R(s) = 5/s$

$\frac{5}{s} \cdot \frac{1}{(s-2)^2} = \frac{5}{s(s-2)^2}$

$\frac{5}{s(s-2)^2} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$

$s=0,\ 5 = A(s-2)^2 + Bs(s-2) + Cs$

$5 = A4$

$A = 5/4$

$s=2,\ 5 = A(s-2)^2 + Bs(s-2) + Cs$

$5 = 2C \to C = 5/2$

$s=1,\ 5 = A(s-2)^2 + Bs(s-2) + Cs$

$5 = 5/4(-1)^2 + B(-1) + 5/2(1)$

$5 = 5/4 - B + 5/2$

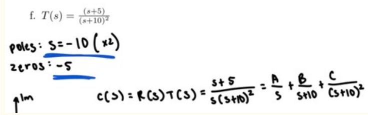
$5 = 1.25 - B + 2.5$

$5/4 = -B,\ B = -5/4$

$= \frac{5}{4s} - \frac{5}{4(s-2)} + \frac{5}{2(s-2)^2}$

$5/4 - 5/4\,e^{2t} + 5/2\,t\,e^{2t}$

amplitude
2.5
time (seconds)
5

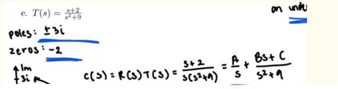


f. $T(s) = \frac{(s+5)}{(s+10)^2}$

poles: $s = -10\ (\times 2)$

zeros: -5

$C(s) = R(s)T(s) = \frac{s+5}{s(s+10)^2} = \frac{A}{s} + \frac{B}{s+10} + \frac{C}{(s+10)^2}$



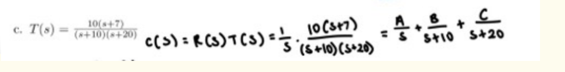
e. $T(s) = \frac{s+2}{s^2+9}$

poles: $\pm 3i$

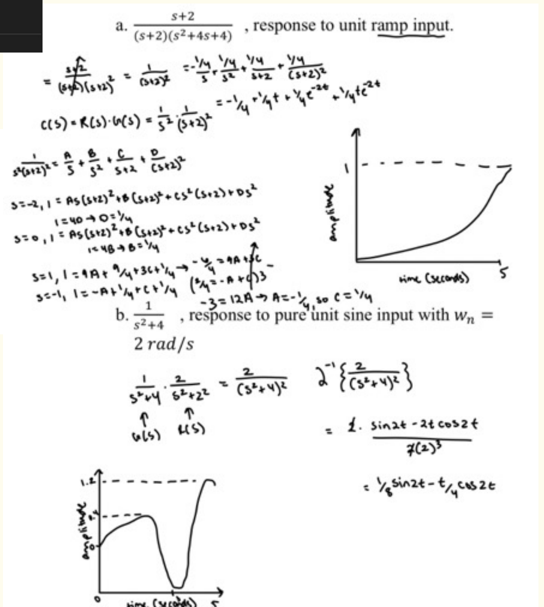
zeros: -2

$C(s) = R(s)T(s) = \frac{s+2}{s(s^2+9)} = \frac{A}{s} + \frac{Bs+C}{s^2+9}$

on und



c. $T(s) = \frac{10(s+7)}{(s+10)(s+20)}$ $C(s) = R(s)T(s) = \frac{1}{s} \cdot \frac{10(s+7)}{(s+10)(s+20)} = \frac{A}{s} + \frac{B}{s+10} + \frac{C}{s+20}$



a. $\frac{s+2}{(s+2)(s^2+4s+4)}$, response to unit ramp input.

$= \frac{\cancel{s+2}}{\cancel{(s+2)}(s+2)^2} = \frac{1}{(s+2)^2}$ $= -\frac{1/4}{s} + \frac{1/4}{s^2} + \frac{1/4}{s+2} + \frac{1/4}{(s+2)^2}$

$= -1/4 + 1/4\,t + 1/4\,e^{-2t} + 1/4\,te^{-2t}$

$C(s) = R(s) \cdot G(s) = \frac{1}{s^2} \cdot \frac{1}{(s+2)^2}$

$\frac{1}{s^2(s+2)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2}$

$s=-2,\ 1 = As(s+2)^2 + B(s+2)^2 + Cs^2(s+2) + Ds^2$

$1 = 4D \to D = 1/4$

$s=0,\ 1 = As(s+2)^2 + B(s+2)^2 + Cs^2(s+2) + Ds^2$

$1 = 4B \to B = 1/4$

$s=1,\ 1 = 9A + 9/4 + 3C + 1/4 \to -\frac{6}{4} = 9A + 3C$

$s=-1,\ 1 = -A + 1/4 + C + 1/4$ $\left(\frac{2}{4} = -A + C\right)3$

$-3 = 12A \to A = -1/4$, so $C = 1/4$

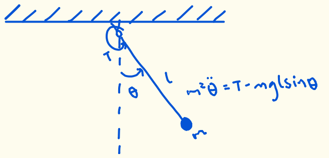
b. $\frac{1}{s^2+4}$, response to pure unit sine input with $w_n = 2\,rad/s$

$\frac{1}{s^2+4} \cdot \frac{2}{s^2+2^2} = \frac{2}{(s^2+4)^2}$ $\mathcal{L}^{-1}\left\{\frac{2}{(s^2+4)^2}\right\}$

$\uparrow$ G(s) $\uparrow$ R(s)

$= 1 \cdot \frac{\sin 2t - 2t\cos 2t}{2(2)^3}$

$= 1/8 \sin 2t - t/4 \cos 2t$



poles of G(s) are in right side of plane, unstable, so no finite value

control → follow, another traffic scenario, redundant

a) $\ddot{y}(t) = 9\dot{y}(t) - 12y(t) + \iint_{0-}^{t} y(t)\,dt^2 + 21u(t)$

$x_1 = \iint_0^t y(t)\,dt^2$

$\dot{x}_1 = x_2 = \int_{-0}^{t} y(t)\,dt$

$\dot{x}_2 = x_3 = y(t)$

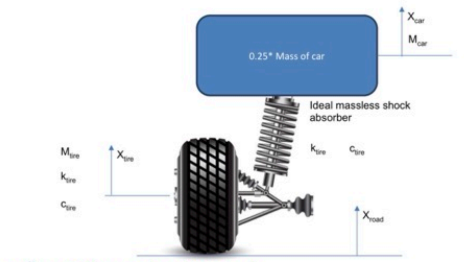
$\dot{x}_3 = x_4 = \dot{y}(t)$

$\dot{x}_4 = \ddot{y}(t) = 9\dot{y}(t) - 12y(t) + \iint_0^t y(t)\,dt^2 + 21u(t)$

b) $\dot{x}_4 = 9x_4 - 12x_3 + x_1 + 21u(t)$

$$A \rightarrow \begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\end{bmatrix} = \begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&-12&9\end{bmatrix},\ B = \begin{bmatrix}0\\0\\0\\21\end{bmatrix}$$

$x_3 = y(t)$ $\quad c = [0\ 0\ 1\ 0],\ D = 0$



$M_c\ddot{X}_{car} = -k_s(X_{car} - X_{tire}) - C_s(\dot{X}_{car} - \dot{X}_{tire})$

$M_t\ddot{X}_{tire} + (C_s + C_t)\dot{X}_{tire} + (k_s + k_t)X_{tire} = k_t X_{road} + C_t\dot{X}_{road}$

$x_1 = X_{car},\ x_2 = \dot{X}_{car},\ x_3 = X_{tire},\ x_4 = \dot{X}_{tire}$

$\dot{x}_1 = x_2,\ \dot{x}_3 = x_4,\ \dot{x}_2 = \frac{-k_s}{M_c}x_1 - \frac{C_s}{M_c}x_2 + \frac{k_s}{M_c}x_3 + \frac{C_s}{M_c}x_4$

$\dot{x}_4 = \frac{k_s}{M_t}x_1 + \frac{C_s}{M_t}x_2 - \frac{(k_s + k_t)}{M_t}x_3 - \frac{(C_s + C_t)}{M_t}x_4 + \frac{k_t}{M_t}X_{road} + \frac{C_t}{M_t}\dot{X}_{road}$

$$A = \begin{bmatrix}0&1&0&0\\ \frac{-k_s}{M_c}&-\frac{C_s}{M_c}&k_s/M_c&\frac{C_s}{M_c}\\ 0&0&0&1\\ k_s/M_t&C_s/M_t&-\frac{k_s+k_t}{M_t}&-\frac{C_s+C_t}{M_t}\end{bmatrix},\ B = \begin{bmatrix}0&0\\0&0\\0&0\\k_t/M_t&C_t/M_t\end{bmatrix},\ C = [1\,0\,0\,0],\ D = 0$$

(5) $\ddot{y}^2(t) + 2\ddot{y}(t)y(t) + y^2(t) = (2u(t))(\ddot{y}(t) + y(t))$ for $u(t) = \sin\omega t$.

$(\ddot{y}(t) + y(t))^{\cancel{2}\,1} = 2u(t)\cancel{(\ddot{y}(t) + y(t))}$

$\ddot{y}(t) + y(t) = 2u(t)$

$\ddot{y}(t) = -y(t) + 2u(t)$

$x_1 = y(t)$

$\dot{x}_1 = x_2 = \dot{y}(t)$

$\dot{x}_2 = x_3 = \ddot{y}(t) = -x_1 + 2u(t)$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0&1\\-1&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} + \begin{bmatrix}0\\2\end{bmatrix}u(t)$$

$$y(t) = [1\ 0]\begin{bmatrix}x_1\\x_2\end{bmatrix} + [0]u(t)$$

$$\ddot{y}(t) = 9\dot{y}(t) - 12\dot{y}(t) + \int_{0-}^{t} y(t)dt + 21\dot{u}(t)$$

$\ddot{y}(t) = 9\dot{y}(t) - 12y(t) + \iint_{0-}^{t} y(t)\,dt^2 + 21u(t)$

$x_1 = \iint_{0-}^{t} y(t)\,dt^2$

$\dot{x}_1 = x_2 = \int_{0-}^{t} y(t)\,dt$

$\dot{x}_2 = x_3 = y(t)$

$\dot{x}_3 = x_4 = \dot{y}(t)$

$\dot{x}_4 = \ddot{y}(t) = 9x_4 - 12x_3 + x_1 + 21u(t)$

$$A = \begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&-12&9\end{bmatrix},\ B = \begin{bmatrix}0\\0\\0\\21\end{bmatrix}$$

$c = [0\ 0\ 1\ 0],\ D = 0$

$\uparrow$
$x_3 = y(t)$

a. (5) Under what condition(s) will the transfer function always respond as a 1st order system

all extra poles and zeros must cancel out so function is exactly 1st order if & only: $\frac{K}{s+p}$

$\{a = d, b = f, c = g\}$ or $\{a = e, b = f, c = g\}$

b. (5) Under what condition can you consider the system to response approximately as a 1st order system

one pole is dominant

$d \ll e,\ d \ll \text{Re}\{\text{roots of } s^2 + fs + g\}$

c. (5) Under what condition will the response of the system to R(s) be always k*R(s) , k being an arbitrary non-zero constant scaling factor, as t -> ∞ while maintaining the same order of magnitude of the original transfer function.

1) All poles must be stable
$d > 0,\ e > 0,\ f > 0,\ g > 0$

2) DC gain must be nonzero & finite
$a \neq 0$ and $C \neq 0$

b) (5) Assuming the mass of the car is infinite, reduce the model to a 2nd order system. Derive the State Space model for the new 2nd order system

$M_c\ddot{X}_{car} = -k_s(X_{car} - X_{tire}) - C_s(\dot{X}_{car} - \dot{X}_{tire})$

$M_t\ddot{X}_{tire} + (C_s + C_t)\dot{X}_{tire} + (k_s + k_t)X_{tire} = k_t X_{road} + C_t\dot{X}_{road}$

↳ then separate X_{tire} + X_{road}

$x_1 = X_{tire},\ x_2 = \dot{X}_{tire},\ \dot{x}_1 = x_2,\ \dot{x}_2 = \ddot{X}_{tire} = -\frac{(k_s + k_t)}{M_t}x_1 - \frac{(C_s + C_t)}{M_t}x_2 + \frac{k_t}{M_t}X_{road} + \frac{C_t}{M_t}\dot{X}_{road}$

$$\begin{bmatrix}0&1\\ \frac{-k_s+k_t}{M_t}&-\frac{C_s+C_t}{M_t}\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} + \begin{bmatrix}0&0\\ \frac{k_t}{M_t}&\frac{C_t}{M_t}\end{bmatrix}\begin{bmatrix}X_{road}\\ \dot{X}_{road}\end{bmatrix}$$

$$y = X_{tire} = [1\ 0]\begin{bmatrix}x_1\\x_2\end{bmatrix}$$

a. (10) Develop the linear model of a just a simple pendulum (vertically down) of length L and mass M with an ideal linear torsional spring K at the pivot joint of the pendulum.

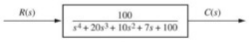
$ML^2\ddot{\theta} + K\theta + MgL\sin\theta = T(t)$ $\quad x_1 = \theta,\ x_2 = \dot{\theta}$

std form $= \ddot{\theta} + \omega_n^2\theta = \frac{1}{ML^2}T(t)$ $\quad \dot{x}_1 = x_2$

$\ddot{\theta} + \frac{K + MgL}{ML^2}\theta = \frac{1}{ML^2}T(t)$ $\quad \dot{x}_2 = \ddot{\theta} = -\frac{K + MgL}{ML^2}x_1 + \frac{1}{ML^2}T(t)$

$$\omega_n = \sqrt{\frac{K + MgL}{ML^2}} \qquad \begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0&1\\ -\frac{K+MgL}{ML^2}&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{ML^2}\end{bmatrix}T(t)$$

$$y = \theta = [1\ 0]\begin{bmatrix}x_1\\x_2\end{bmatrix}$$



$$\frac{C(s)}{R(s)} = \frac{100}{s^4 + 20s^3 + 10s^2 + 7s + 100}$$

$(s^4 + 20s^3 + 10s^2 + 7s + 100)C(s) = 100R(s)$

$c''''(t) + 20\dddot{c}(t) + 10\ddot{c}(t) + 7\dot{c}(t) + 100c(t) = 100r(t)$

$x_1 = c(t)$

$\dot{x}_1 = \dot{c}(t) = x_2$

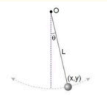
$\dot{x}_2 = \ddot{c}(t) = x_3$

$\dot{x}_3 = \dddot{c}(t) = x_4$

$\dot{x}_4 = c''''(t) = -100x_1 - 7x_2 - 10x_3 - 20x_4 + 100r(t)$

$$A = \begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\-100&-7&-10&-20\end{bmatrix},\ B = \begin{bmatrix}0\\0\\0\\100\end{bmatrix},$$

$C = [1\ 0\ 0\ 0],\ D = 0$



c) (5) Derive the transfer function $\frac{\theta(s)}{\theta_0}$ for the pendulum above, where θ_0 is a non-zero initial angle of the pendulum.

$L\ddot{\theta}(t) = -g\theta(t)$

$L\ddot{\theta}(t) + g\theta(t) = 0$

$\ddot{\theta} + g/L\,\theta = 0$

$\underbrace{s^2\theta(s) - s\theta_0 + g/L\,\theta(s)}{} = 0$

$\frac{\theta(s)}{\theta_0} = \frac{s}{s^2 + g/L}$

d) (5) Under what conditions are the transfer functions in (a) and (c) equivalent.

$\frac{X(s)}{X_0} = \frac{Ms + f_v}{Ms^2 + f_v s + K}$ $\quad$ 1) $f_v = 0$

$\frac{\theta(s)}{\theta_0} = \frac{s}{s^2 + g/L}$ $\quad$ 2) $K/M = g/L = \omega_n^2$

n in the table, determine if the subsystem is lamped, underdamped or undamped. Explain your ubsystem pole locations.

$$\frac{Z_u}{Z_q} = \frac{k_t + C_t s}{m_u s^2 + k_t + C_t s} = \frac{500000\,N/m + 15020\,\frac{N\cdot s}{m}\,s}{320\,kg\,s^2 + 500000\,N/m + 15020\,\frac{N\cdot s}{m}\,s}$$

$320s^2 + 15020s + 500000$

$$s = \frac{-15020 \pm \sqrt{15020^2 - 4(320)(500000)}}{2(320)}$$

$- \pm i$

stable and underdamped

e) (5) Plot the poles, and zeroes (if any) for the total system.

Zeros: $(C_t s + k_t)(C_s s + k_s)$

$s = -\frac{k_s}{C_s} = -228.57$

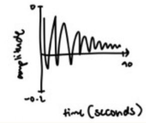
$s = -\frac{k_t}{C_t} = -33.29$

poles:

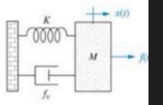
$s^4 + 48.171s^3 + 1851.071s^2 + 1720.75s^5 + 50000 = 0$

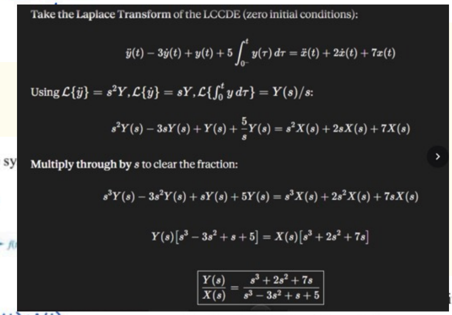
solve for s: $\quad -24 \pm 35i$

$-0.1 + 5i$



(20) a) (5) Derive the transfer function for the sy





Take the Laplace Transform of the LCCDE (zero initial conditions):

$$\ddot{y}(t) - 3\dot{y}(t) + y(t) + 5\int_{0^-}^{t} y(\tau)\,d\tau = \ddot{x}(t) + 2\dot{x}(t) + 7x(t)$$

Using $\mathcal{L}\{\ddot{y}\} = s^2Y, \mathcal{L}\{\dot{y}\} = sY, \mathcal{L}\{\int_0^t y\,d\tau\} = Y(s)/s$:

$$s^2Y(s) - 3sY(s) + Y(s) + \frac{5}{s}Y(s) = s^2X(s) + 2sX(s) + 7X(s)$$

Multiply through by s to clear the fraction:

$$s^3Y(s) - 3s^2Y(s) + sY(s) + 5Y(s) = s^3X(s) + 2s^2X(s) + 7sX(s)$$

$$Y(s)[s^3 - 3s^2 + s + 5] = X(s)[s^3 + 2s^2 + 7s]$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{s^3 + 2s^2 + 7s}{s^3 - 3s^2 + s + 5}}$$

$M\ddot{x}(t) = -f_v\dot{x}(t) - Kx(t) + f(t)$

$Ms^2x(s) = -f_v s x(s) - Kx(s) + f(s)$

$x(s)[Ms^2 + f_v s + K] = f(s)$

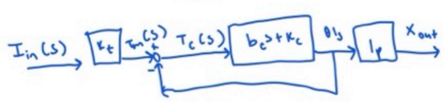
$$\frac{x(s)}{F(s)} = \frac{1}{Ms^2 + f_v s + K}$$

b) (5) Evaluate the final value of X(t) to an initial condition of X₀

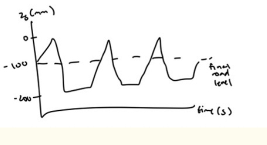
$M[s^2X(s) - sX_0] + f_v[sX(s) - X_0] + KX(s) = 0$

$$X(s) = \frac{X_0(Ms + f_v)}{Ms^2 + f_v s + K}$$

$$x(\infty) = \lim_{s\to 0} sX(s) = \lim_{s\to 0}\frac{sX_0(Ms + f_v)}{Ms^2 + f_v s + K} = \frac{0}{K} = 0$$



g) (5) Plot the total response of the car (Z_s) to a 100 mm drop intc pothole.

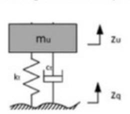


(5) Based on the poles, what can you say about the characteristics of the full response of the system to the same drop into the pothole.

when tire hits pothole (t=0), tire drops rapidly → high-freq ripple, b/c real part is so heavily negative (-24.01), initial vibration is completely choked out and disappears

the other dominates what the passengers feel. since they real part (-0.071), that means main suspension damper is extremely weak

ive the transfer function $\frac{Z_u}{Z_q}$ for only the wheel and tire subsystem **without** the mass of the Body, or the spring and damper of the suspension subsystem. Thus, the transfer function of only



$\Sigma F = m_u\ddot{z}_u$

$n = mg \cdot m/s^2$

$m_u\ddot{z}_u = -k_t(z_u - z_q) - C_t(\dot{z}_u - \dot{z}_q)$

$m_u z_u s^2 = -k_t(z_u - z_q) - C_t s(z_u - z_q)$

$m_u s^2 z_u + k_t z_u - \underline{k_t z_q} + C_t s z_u - \underline{C_t s z_q} = 0$

$z_u(m_u s^2 + k_t + C_t s) = z_q(k_t + C_t s)$

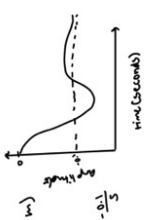
$$\frac{z_u}{z_q} = \frac{k_t + C_t s}{m_u s^2 + k_t + C_t s}$$

p into a n Z_q

$$\frac{Z_u}{Z_q} = \frac{500000\,N/m + 15020\,\frac{N\cdot s}{m}\,s}{320\,kg\,s^2 + 500000\,N/m + 15020\,\frac{N\cdot s}{m}\,s} = \frac{15020s + 500000}{320s^2 + 15020s + 500000}$$

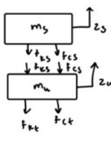
sudden drop into 100mm (0.1m) is represented mathematically as a negative step unit

$z_q(t) = -0.1 \cdot u(t) \Rightarrow Z_q(s) = \frac{-0.1}{s}$



d) (20) Derive the full transfer function for the car suspension system Z_q.

Include a Free Body Diagram in your derivation.



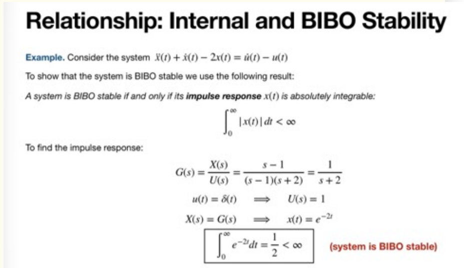
$\rightarrow m_s\ddot{z}_s = -k_s(z_s - z_u) - C_s(\dot{z}_s - \dot{z}_u)$

$$z_u = \frac{m_s s^2 + C_s s + k_s}{C_s s + k_s}$$

$\rightarrow m_u\ddot{z}_u = -k_s(z_u - z_s) - C_s(\dot{z}_u - \dot{z}_s) - k_t(z_u - z_q) - C_t(\dot{z}_u - \dot{z}_q)$

ok, then combine both eqns

$$\frac{Z_s(s)}{Z_q(s)} = \frac{(C_t s + k_t)(C_s s + k_s)}{M_s m_u s^4 + [m_s(C_s + C_t) + m_u C_s]s^3 + [m_s(k_s + k_t) + m_u k_s + C_s C_t]s^2}$$



Relationship: Internal and BIBO Stability

Example. Consider the system $\ddot{x}(t) + \dot{x}(t) - 2x(t) = \dot{u}(t) - u(t)$

To show that the system is BIBO stable we use the following result:

*A system is BIBO stable if and only if its **impulse response** $x(t)$ is absolutely integrable:*

$$\int_0^\infty |x(t)|\,dt < \infty$$

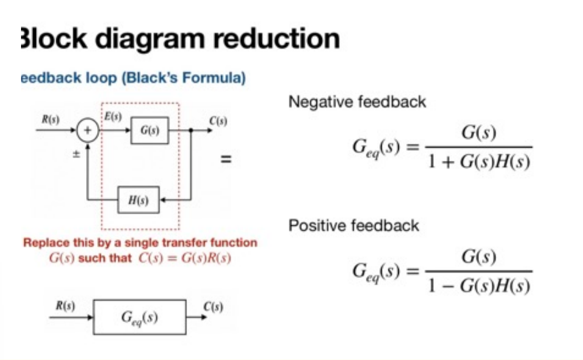
To find the impulse response:

$$G(s) = \frac{X(s)}{U(s)} = \frac{s-1}{(s-1)(s+2)} = \frac{1}{s+2}$$

$$u(t) = \delta(t) \implies U(s) = 1$$

$$X(s) = G(s) \implies x(t) = e^{-2t}$$

$$\boxed{\int_0^\infty e^{-2t}dt = \frac{1}{2} < \infty}$$ (system is BIBO stable)



3lock diagram reduction

eedback loop (Black's Formula)

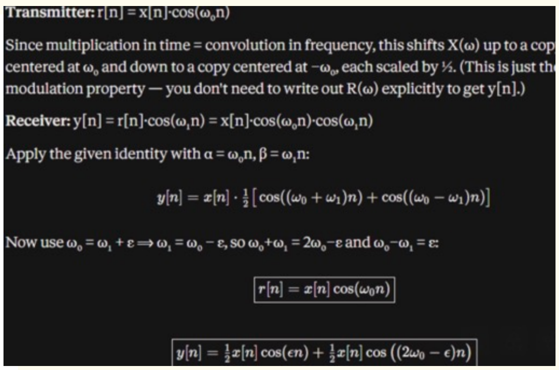
Replace this by a single transfer function $G(s)$ such that $C(s) = G(s)R(s)$

Negative feedback

$$G_{eq}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Positive feedback

$$G_{eq}(s) = \frac{G(s)}{1 - G(s)H(s)}$$



Transmitter: r[n] = x[n]·cos(ω₀n)

Since multiplication in time = convolution in frequency, this shifts X(ω) up to a cop centered at ω₀ and down to a copy centered at −ω₀, each scaled by ½. (This is just th modulation property — you don't need to write out R(ω) explicitly to get y[n].)

Receiver: y[n] = r[n]·cos(ω₁n) = x[n]·cos(ω₀n)·cos(ω₁n)

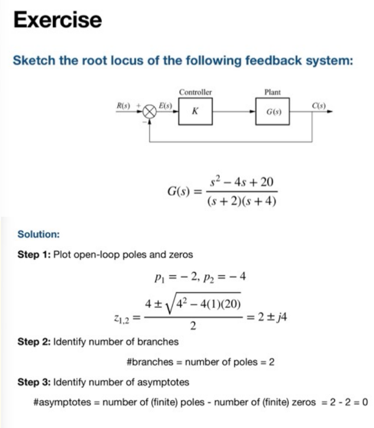
Apply the given identity with α = ω₀n, β = ω₁n:

$$y[n] = x[n] \cdot \tfrac{1}{2}\left[\cos((\omega_0 + \omega_1)n) + \cos((\omega_0 - \omega_1)n)\right]$$

Now use ω₀ = ω₁ + ε ⟹ ω₁ = ω₀ − ε, so ω₀+ω₁ = 2ω₀−ε and ω₀−ω₁ = ε:

$$\boxed{r[n] = x[n]\cos(\omega_0 n)}$$

$$\boxed{y[n] = \tfrac{1}{2}x[n]\cos(\epsilon n) + \tfrac{1}{2}x[n]\cos\left((2\omega_0 - \epsilon)n\right)}$$



Exercise

Sketch the root locus of the following feedback system:

$$G(s) = \frac{s^2 - 4s + 20}{(s+2)(s+4)}$$

Solution:

Step 1: Plot open-loop poles and zeros

$$p_1 = -2,\ p_2 = -4$$

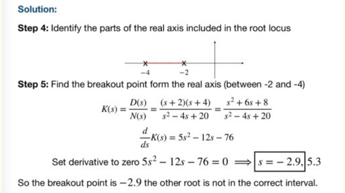
$$z_{1,2} = \frac{4 \pm \sqrt{4^2 - 4(1)(20)}}{2} = 2 \pm j4$$

Step 2: Identify number of branches

#branches = number of poles = 2

Step 3: Identify number of asymptotes

#asymptotes = number of (finite) poles - number of (finite) zeros = 2 - 2 = 0



Solution:

Step 4: Identify the parts of the real axis included in the root locus

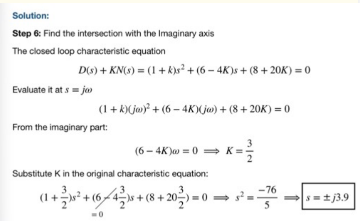
Step 5: Find the breakout point form the real axis (between -2 and -4)

$$K(s) = \frac{D(s)}{N(s)} = \frac{(s+2)(s+4)}{s^2 - 4s + 20} = \frac{s^2 + 6s + 8}{s^2 - 4s + 20}$$

$$\frac{d}{ds}K(s) = 5s^2 - 12s - 76$$

Set derivative to zero $5s^2 - 12s - 76 = 0 \implies \boxed{s = -2.9,}\,5.3$

So the breakout point is -2.9 the other root is not in the correct interval.



Solution:

Step 6: Find the intersection with the Imaginary axis

The closed loop characteristic equation

$$D(s) + KN(s) = (1+k)s^2 + (6-4K)s + (8+20K) = 0$$

Evaluate it at $s = j\omega$

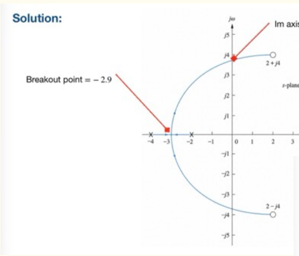
$$(1+k)(j\omega)^2 + (6-4K)(j\omega) + (8+20K) = 0$$

From the imaginary part:

$$(6-4K)\omega = 0 \implies K = \frac{3}{2}$$

Substitute K in the original characteristic equation:

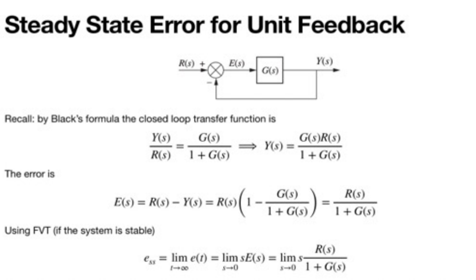
$$(1+\frac{3}{2})s^2 + \underbrace{(6 \cancel{-} 4\frac{3}{2})}_{=0}s + (8+20\frac{3}{2}) = 0 \implies s^2 = \frac{-76}{5} \implies \boxed{s = \pm j3.9}$$



Solution:

Im axis crossing = 3.9

Breakout point = − 2.9



Steady State Error for Unit Feedback

Recall: by Black's formula the closed loop transfer function is

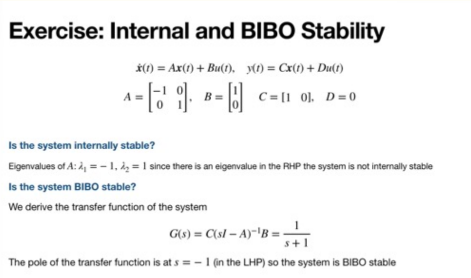
$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} \implies Y(s) = \frac{G(s)R(s)}{1+G(s)}$$

The error is

$$E(s) = R(s) - Y(s) = R(s)\left(1 - \frac{G(s)}{1+G(s)}\right) = \frac{R(s)}{1+G(s)}$$

Using FVT (if the system is stable)

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{R(s)}{1+G(s)}$$



Exercise: Internal and BIBO Stability

$$\dot{x}(t) = Ax(t) + Bu(t),\quad y(t) = Cx(t) + Du(t)$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix},\quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}\quad C = [1\ \ 0],\quad D = 0$$

Is the system internally stable?

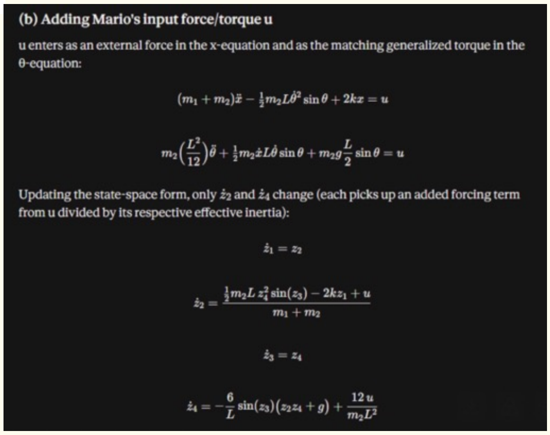
Eigenvalues of A: $\lambda_1 = -1,\ \lambda_2 = 1$ since there is an eigenvalue in the RHP the system is not internally stable

Is the system BIBO stable?

We derive the transfer function of the system

$$G(s) = C(sI - A)^{-1}B = \frac{1}{s+1}$$

The pole of the transfer function is at $s = -1$ (in the LHP) so the system is BIBO stable



(b) Adding Mario's input force/torque u

u enters as an external force in the x-equation and as the matching generalized torque in the θ-equation:

$$(m_1 + m_2)\ddot{x} - \tfrac{1}{2}m_2 L\dot{\theta}^2\sin\theta + 2kx = u$$

$$m_2\left(\frac{L^2}{12}\right)\ddot{\theta} + \tfrac{1}{2}m_2\dot{x}L\dot{\theta}\sin\theta + m_2 g\frac{L}{2}\sin\theta = u$$

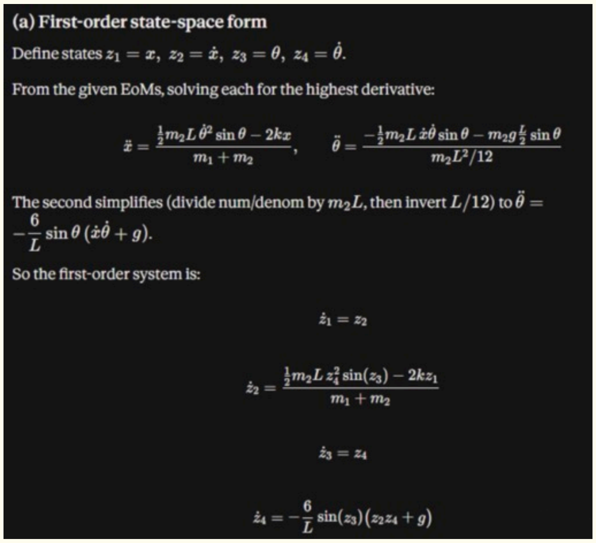
Updating the state-space form, only $\dot{z}_2$ and $\dot{z}_4$ change (each picks up an added forcing term from u divided by its respective effective inertia):

$$\dot{z}_1 = z_2$$

$$\dot{z}_2 = \frac{\tfrac{1}{2}m_2 L\, z_4^2 \sin(z_3) - 2kz_1 + u}{m_1 + m_2}$$

$$\dot{z}_3 = z_4$$

$$\dot{z}_4 = -\frac{6}{L}\sin(z_3)(z_2 z_4 + g) + \frac{12\,u}{m_2 L^2}$$



(a) First-order state-space form

Define states $z_1 = x,\ z_2 = \dot{x},\ z_3 = \theta,\ z_4 = \dot{\theta}$.

From the given EoMs, solving each for the highest derivative:

$$\ddot{x} = \frac{\tfrac{1}{2}m_2 L\,\dot{\theta}^2\sin\theta - 2kx}{m_1 + m_2},\qquad \ddot{\theta} = \frac{-\tfrac{1}{2}m_2 L\,\dot{x}\dot{\theta}\sin\theta - m_2 g\tfrac{L}{2}\sin\theta}{m_2 L^2/12}$$

The second simplifies (divide num/denom by $m_2 L$, then invert $L/12$) to $\ddot{\theta} = -\frac{6}{L}\sin\theta\,(\dot{x}\dot{\theta} + g)$.

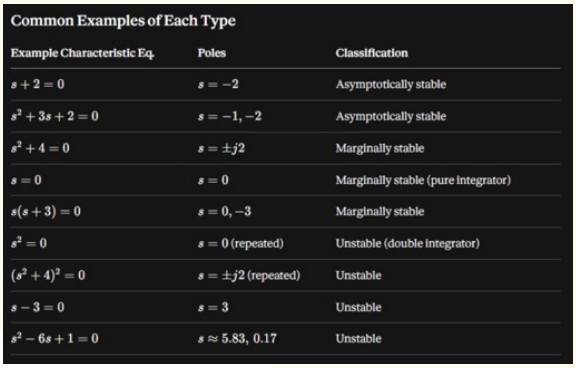
So the first-order system is:

$$\dot{z}_1 = z_2$$

$$\dot{z}_2 = \frac{\tfrac{1}{2}m_2 L\, z_4^2\sin(z_3) - 2kz_1}{m_1 + m_2}$$

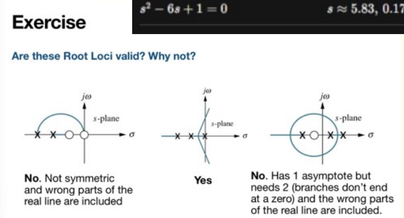
$$\dot{z}_3 = z_4$$

$$\dot{z}_4 = -\frac{6}{L}\sin(z_3)(z_2 z_4 + g)$$



Common Examples of Each Type

Example Characteristic Eq.	Poles	Classification
$s+2=0$	$s=-2$	Asymptotically stable
$s^2+3s+2=0$	$s=-1,-2$	Asymptotically stable
$s^2+4=0$	$s=\pm j2$	Marginally stable
$s=0$	$s=0$	Marginally stable (pure integrator)
$s(s+3)=0$	$s=0,-3$	Marginally stable
$s^2=0$	$s=0$ (repeated)	Unstable (double integrator)
$(s^2+4)^2=0$	$s=\pm j2$ (repeated)	Unstable
$s-3=0$	$s=3$	Unstable
$s^2-6s+1=0$	$s\approx 5.83,\ 0.17$	Unstable



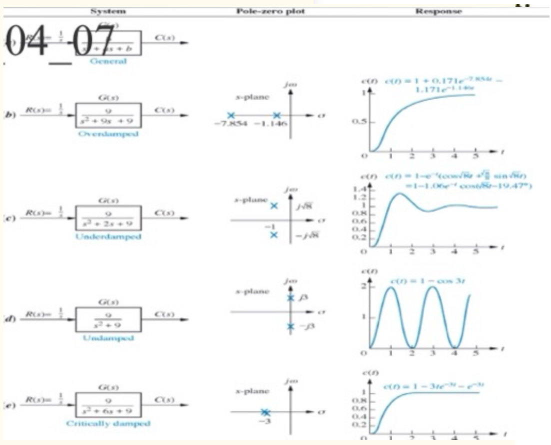
Exercise

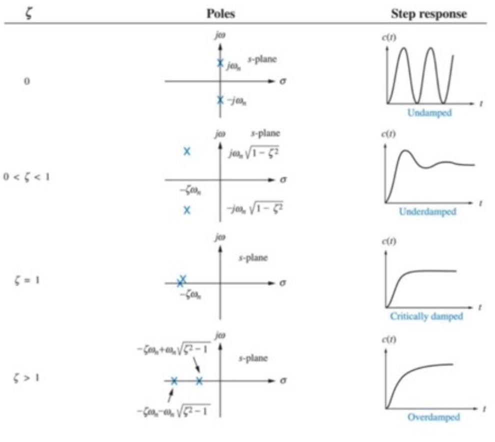
Are these Root Loci valid? Why not?

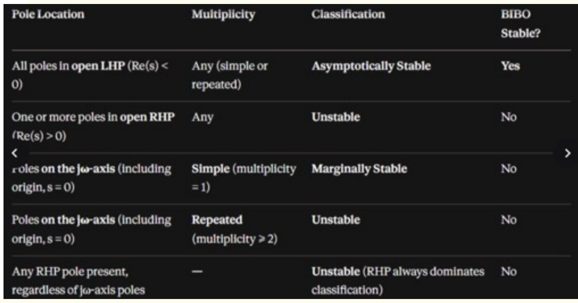
No. Not symmetric and wrong parts of the real line are included

Yes

No. Has 1 asymptote but needs 2 (branches don't end at a zero) and the wrong parts of the real line are included.







Pole Location	Multiplicity	Classification	BIBO Stable?
All poles in **open LHP** (Re(s) < 0)	Any (simple or repeated)	**Asymptotically Stable**	**Yes**
One or more poles in **open RHP** (Re(s) > 0)	Any	**Unstable**	No
Poles **on the jω-axis** (including origin, s = 0)	**Simple** (multiplicity = 1)	**Marginally Stable**	No
Poles **on the jω-axis** (including origin, s = 0)	**Repeated** (multiplicity ≥ 2)	**Unstable**	No
Any RHP pole present, regardless of jω-axis poles	—	**Unstable** (RHP always dominates classification)	No

4. (10) Express each system below in Standard State Space form and find the final value for each output y(t) subjected to a unit step input u(t). (2 points each)

$\rightarrow s^2Y(s)-6sY(s)+Y(s)=U(s)\rightarrow Y(s)=\frac{\overset{1/s}{U(s)}}{s^2-6s+1}=\frac{1/s}{s^2-6s+1}=\frac{1}{s\underbrace{(s^2-6s+1)}_{s=5.83,\,0.17}}$ so both pos. real, FVT fails so infinit

a. $\ddot{y}(t) - 6\dot{y}(t) + y(t) = u(t)$

$\ddot{y}(t)=6\dot{y}(t)-y+u(t)$

let $x_1=y$, $x_2=\dot{y}$

$\dot{x}_1=\dot{y}=x_2$

$\dot{x}_2=\ddot{y}=-x_1+6x_2+u$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0&1\\-1&6\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}+\begin{bmatrix}0\\1\end{bmatrix}u,\quad y=\begin{bmatrix}1&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}$$

b. $\ddot{y}(t) + 2y(t) + 1 = u(t)$

$\ddot{y}=-2y-1+u(t)$

$x_1=y,\ x_2=\dot{y}\rightarrow \dot{x}_1=\dot{y}=x_2$

$\dot{x}_2=\ddot{y}=-2x_1-1+u$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0&1\\-2&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}+\begin{bmatrix}0\\-1\end{bmatrix}u+\begin{bmatrix}0\\-1\end{bmatrix},\ y=\begin{bmatrix}1&0\end{bmatrix}x$$

$s^2Y(s)+2Y(s)+1/s=\underset{\uparrow 1/s}{U(s)}\rightarrow s^2Y(s)+2Y(s)=1/s-1/s=0$

$Y(s)[s^2+2]=0\rightarrow Y(s)=0\rightarrow y(\infty)=0$

FVT is valid if and only if every pole of sY(s) lies in the open, left-half-plane, w/ at most one exception: a single, simple (non-repeated) pole at the origin s=0

c. $u(t) = 18(\dot{y}(t) + 6y(t))$

$\dot{y}(t)=-6y(t)+1/18\,u(t)$

$x_1=y\rightarrow \dot{x}_1=-6x_1+1/18\,u$

$A=-6,\ B=1/18,\ C=1,\ D=0$

$sY(s)=-6Y(s)+1/18\,U(s)\rightarrow Y(s)=\frac{1}{18s(s+6)}\rightarrow y(\infty)=\lim_{s\to0}s\cdot Y(s)=\lim_{s\to0}\frac{s}{18(s+6)}=\frac{1}{108}$

d. $\ddot{y}(t) = 13 - \dot{y}(t) - 4y(t) + u(t)$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0&1\\-4&-1\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}+\begin{bmatrix}0\\1\end{bmatrix}u+\begin{bmatrix}0\\13\end{bmatrix},\ y=\begin{bmatrix}1&0\end{bmatrix}x$$

$Y(s)=\frac{14}{s(s^2+s+4)}\rightarrow s=-0.5\pm j1.94$

$y(\infty)=\lim_{s\to0}s\cdot Y(s)=\lim_{s\to0}\frac{14}{s^2+s+4}=\frac{14}{4}$

e. $\dot{y}(t) = -u(t)$

$x_1=y\rightarrow \dot{x}_1=-u(t),\ y=x_1$

$A=0,\ B=-1,\ C=1,\ D=0$

$sY(s)=-U(s)\rightarrow Y(s)=-\frac{U(s)}{s}=-\frac{1}{s}\cdot\frac{1}{s}=-\frac{1}{s^2}\rightarrow sY(s)=-\frac{1}{s}\rightarrow \lim_{s\to0}sY(s)=-\frac{1}{s}=\infty$

a. $te^{-4t} = \ddot{y}(t) - 6\dot{y}(t) + y(t)$

$=s^2-6s+1=0$

$s=\frac{6\pm\sqrt{36-4}}{2}=3\pm2\sqrt{2}$ ← both real, positive, lie in RHP, unstable

b. $\underbrace{\ddot{y}(t) + 2y(t)}_{} = (2e^{4t}\sin(wt) - 1)$

s^2+2

$s=\pm j\sqrt{2}\rightarrow$ marginally stable, if repeated such as $(s^2+2)^2=0$, unstable

c. $u(t) = 18(\dot{y}(t) - 6)$

$\dot{y}(t)-\frac{1}{18}u(t)=6$

$18s=0$

$s=0$, pole at origin, not repeated → marginally stable

d. $\dddot{y}(t) = 13 - \dot{y}(t)$

$s^3+s=0$

$s=0,i,i\rightarrow$ unstable

e. $\dot{y}(t) = \sin\omega t$

$\dot{y}(t)=0\rightarrow s=0$, so marginally stable

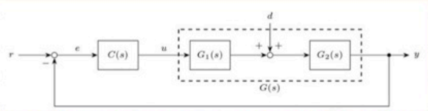
↑ as long as $w\neq0$, system is stable, if $w=0$ the system is in steady state

Feedback control systems:

- use output to tune input
- help a system follow a desired reference
- compare actual output w/ desired input and minimize the error
- everywhere / everyday systems

shower knob – inputs: heat and pressure setting of the knob; outputs: perceived temp + pressure

speed controller – has accurate measurement of current speed and accurate actuation of pedal position – feedback controller



$e=r-y;\ u=C(s)e,\ y=G_2(s)[G_1(s)u+d(s)]$

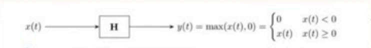
$y=C(s)G_1(s)G_2(s)e(s)+G_2(s)d(s)=C(s)G(s)[r(s)-y(s)]+G_2(s)d(s)$

$y(s)[1+C(s)G(s)]=C(s)G(s)r(s)+G_2(s)d(s)$

$y(s)=\frac{C(s)G(s)}{1+C(s)G(s)}r(s)+\frac{G_2(s)}{1+C(s)G(s)}d(s)$

$G(s)=\frac{s+4}{(s-6)(s-2)}\rightarrow$ does not approach 1/3 as FVT can't be applied since poles are on right-side plane

state-space are never unique. any similarity transformation $\tilde{x}=Tx$ produces a diff. (A,B,C,D) w/ same transfer func.



$y(t)=\max(x(t),0)=\begin{cases}0 & x(t)<0\\ x(t) & x(t)\geq0\end{cases}$

↳ can't be linear:

let $x_1(t)=1,\ x_2(t)=-1$

$H(x_1)=\max(1,0)=1$

$H(x_2)=\max(-1,0)=0$

$H(x_1)+H(x_2)=1$, but $x_1(t)+x_2(t)=0$,

so $H(x_1+x_2)=\max(0,0)=0\neq1$

4.3]

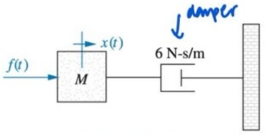


FIGURE P4.3

FBD on M:

$M\ddot{x}=f(t)-b\dot{x}$

$f(t)=M\dot{v}+bv$

now laplace transform:

$F(s)=MsV(s)+bV(s)$

$F(s)=V(s)[Ms+b]$

$\frac{V(s)}{F(s)}=\frac{1}{Ms+b}$

now to match standard $\frac{K}{s+a}$ form:

$\frac{V(s)}{F(s)}=\frac{1/M}{s+b/M}$ so $a=b/m$ and since $b=6\,NS/m$

$a=6/M$

a) settling time eqn. for first-order system:

$T_s=4/a$

$T_s=4/(6/M)=\frac{4M}{6}=\underline{\frac{2M}{3}\text{ seconds}}$

b) Rise time eqn. for first-order system.

$T_r=\frac{2.2}{a}$

$T_r=\frac{2.2}{6/M}=\frac{2.2M}{6}\approx\underline{0.367M\text{ seconds}}$

he following conditions (*Chiu, 1997*). [Sec: 3.5]

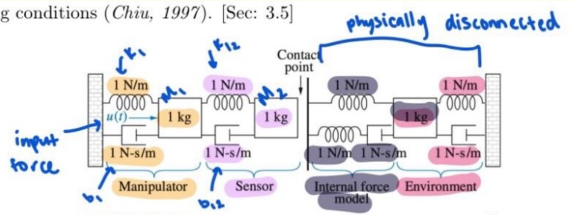


FIGURE P3.14 Robotic manipulator and target environment[11]

a. The manipulator is not in contact with its target environment.

FBD (f=ma)

$$\underbrace{m_1}_{N}\ddot{x}_1 = u(t) - K_1x_1 - b_1\dot{x}_1 - K_{12}(x_1-x_2) - b_{12}(\dot{x}_1-\dot{x}_2)$$

$$m_2\ddot{x}_2 = K_{12}(x_1-x_2) + b_{12}(\dot{x}_1-\dot{x}_2)$$

$$\ddot{x}_1 = u - x_1 - \dot{x}_1 - 1(x_1-x_2) - 1(\dot{x}_1-\dot{x}_2)$$

$$\ddot{x}_1 = u - x_1 - \dot{x}_1 - x_1 + x_2 - \dot{x}_1 + \dot{x}_2$$

$$\ddot{x}_1 = u - 2x_1 - 2\dot{x}_1 + x_2 + \dot{x}_2$$

$$\ddot{x}_2 = 1(x_1-x_2) + 1(\dot{x}_1-\dot{x}_2)$$

$$\ddot{x}_2 = x_1 - x_2 + \dot{x}_1 - \dot{x}_2$$

b. The manipulator is in constant contact with its target environment.

FBD (f=ma) and using that m=1, K=1, and b=1

$$\ddot{x}_1 = u - 2x_1 - 2\dot{x}_1 + x_2 + \dot{x}_2$$

$$\ddot{x}_2 = x_1 + \dot{x}_1 - 2x_2 - \dot{x}_2 + x_3$$

$$\ddot{x}_3 = x_2 - 2x_3 - \dot{x}_3 + x_4 + \dot{x}_4$$

$$\ddot{x}_4 = x_3 - x_4 + \dot{x}_3 - 2\dot{x}_4$$

so let:

$$z_1 = x_1,\ z_2 = \dot{x}_1,\ z_3 = x_2,\ z_4 = \dot{x}_2,\ z_5 = x_3,\ z_6 = \dot{x}_3,\ x_7 = x_4,\ z_8 = \dot{x}_4$$

$$G(s) = \frac{C(s)}{R(s)} = \frac{b_{n-1}s^{n-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

↑ transfer function

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix},\quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

$C = [b_0\ b_1\ b_2 \dots b_{n-1}]$ ← phase-variable form

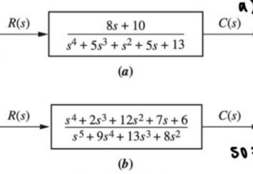

R(s)
$\frac{8s+10}{s^4+5s^3+s^2+5s+13}$
C(s)
(a)
R(s)
$\frac{s^4+2s^3+12s^2+7s+6}{s^5+9s^4+13s^3+8s^2}$
C(s)
(b)

FIGURE P3.9

a) n = 4, so 4 state variables

$a_0 = 13$ and $b_0 = 10$
$a_1 = 5$ $b_1 = 8$
$a_2 = 1$ $b_2 = 0$
$a_3 = 5$ $b_3 = 0$

so:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -13 & -5 & -1 & -5 \end{bmatrix},\quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

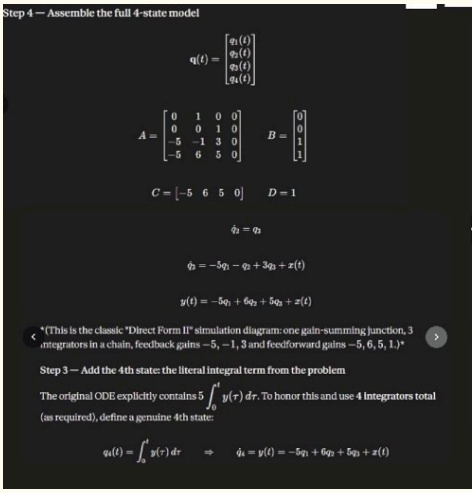
$C = [10\ 8\ 0\ 0]$, $D = 0$

and state equations are

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = x_4$$
$$\dot{x}_4 = -13x_1 - 5x_2 - x_3 - 5x_4 + r(t)$$

and output equation is:

$$y(t) = 10x_1 + 8x_2$$



Step 4 — Assemble the full 4-state model

$$\mathbf{q}(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \\ q_4(t) \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -5 & -1 & 3 & 0 \\ -5 & 6 & 5 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -5 & 6 & 5 & 0 \end{bmatrix} \qquad D = 1$$

$$\dot{q}_2 = q_3$$

$$\dot{q}_3 = -5q_1 - q_2 + 3q_3 + x(t)$$

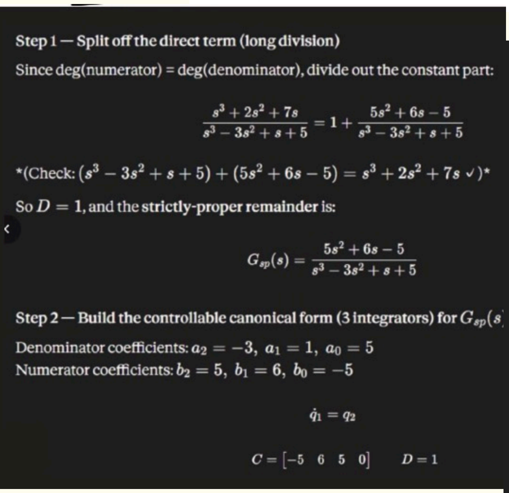
$$y(t) = -5q_1 + 6q_2 + 5q_3 + x(t)$$

(This is the classic "Direct Form II" simulation diagram: one gain-summing junction, 3 integrators in a chain, feedback gains $-5, -1, 3$ and feedforward gains $-5, 6, 5, 1$.)

Step 3 — Add the 4th state: the literal integral term from the problem

The original ODE explicitly contains $5\int_0^t y(\tau)\,d\tau$. To honor this and use **4 integrators total** (as required), define a genuine 4th state:

$$q_4(t) = \int_0^t y(\tau)\,d\tau \quad\Rightarrow\quad \dot{q}_4 = y(t) = -5q_1 + 6q_2 + 5q_3 + x(t)$$



Step 1 — Split off the direct term (long division)

Since deg(numerator) = deg(denominator), divide out the constant part:

$$\frac{s^3+2s^2+7s}{s^3-3s^2+s+5} = 1 + \frac{5s^2+6s-5}{s^3-3s^2+s+5}$$

(Check: $(s^3-3s^2+s+5)+(5s^2+6s-5) = s^3+2s^2+7s$ ✓)

So $D = 1$, and the **strictly-proper remainder** is:

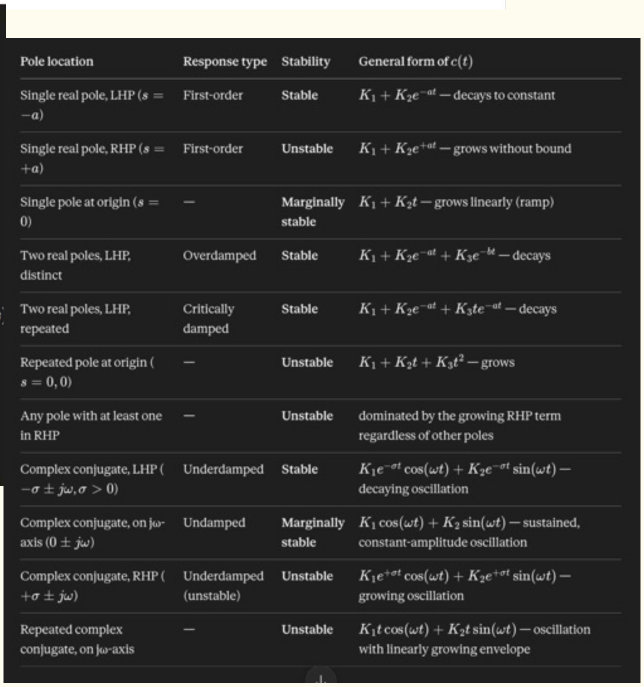
$$G_{sp}(s) = \frac{5s^2+6s-5}{s^3-3s^2+s+5}$$

Step 2 — Build the controllable canonical form (3 integrators) for $G_{sp}(s$

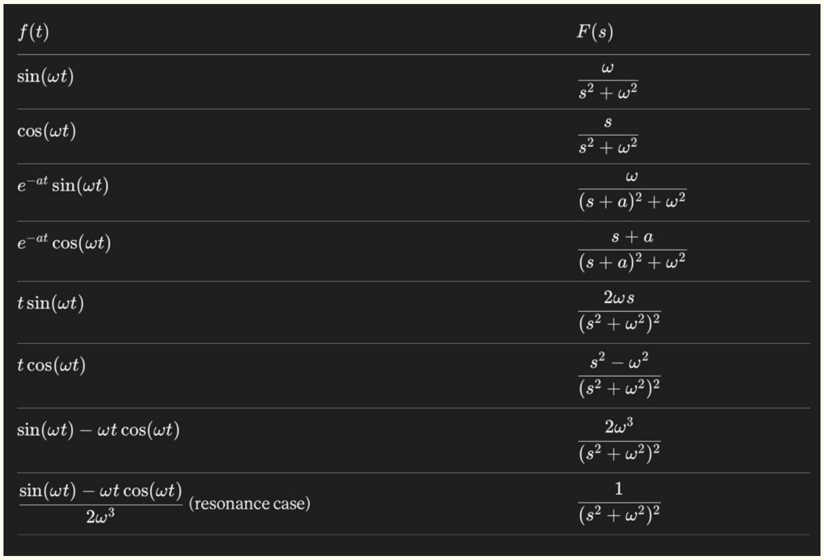
Denominator coefficients: $a_2 = -3,\ a_1 = 1,\ a_0 = 5$
Numerator coefficients: $b_2 = 5,\ b_1 = 6,\ b_0 = -5$

$$\dot{q}_1 = q_2$$

$$C = \begin{bmatrix} -5 & 6 & 5 & 0 \end{bmatrix} \qquad D = 1$$



Pole location	Response type	Stability	General form of $c(t)$
Single real pole, LHP ($s = -a$)	First-order	Stable	$K_1 + K_2e^{-at}$ — decays to constant
Single real pole, RHP ($s = +a$)	First-order	Unstable	$K_1 + K_2e^{+at}$ — grows without bound
Single pole at origin ($s = 0$)	—	Marginally stable	$K_1 + K_2t$ — grows linearly (ramp)
Two real poles, LHP, distinct	Overdamped	Stable	$K_1 + K_2e^{-at} + K_3e^{-bt}$ — decays
Two real poles, LHP, repeated	Critically damped	Stable	$K_1 + K_2e^{-at} + K_3te^{-at}$ — decays
Repeated pole at origin ($s = 0, 0$)	—	Unstable	$K_1 + K_2t + K_3t^2$ — grows
Any pole with at least one in RHP	—	Unstable	dominated by the growing RHP term regardless of other poles
Complex conjugate, LHP ($-\sigma \pm j\omega, \sigma > 0$)	Underdamped	Stable	$K_1e^{-\sigma t}\cos(\omega t) + K_2e^{-\sigma t}\sin(\omega t)$ — decaying oscillation
Complex conjugate, on $j\omega$-axis ($0 \pm j\omega$)	Undamped	Marginally stable	$K_1\cos(\omega t) + K_2\sin(\omega t)$ — sustained, constant-amplitude oscillation
Complex conjugate, RHP ($+\sigma \pm j\omega$)	Underdamped (unstable)	Unstable	$K_1e^{+\sigma t}\cos(\omega t) + K_2e^{+\sigma t}\sin(\omega t)$ — growing oscillation
Repeated complex conjugate, on $j\omega$-axis	—	Unstable	$K_1t\cos(\omega t) + K_2t\sin(\omega t)$ — oscillation with linearly growing envelope



$f(t)$	$F(s)$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$
$t\sin(\omega t)$	$\frac{2\omega s}{(s^2+\omega^2)^2}$
$t\cos(\omega t)$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
$\sin(\omega t) - \omega t\cos(\omega t)$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$
$\frac{\sin(\omega t) - \omega t\cos(\omega t)}{2\omega^3}$ (resonance case)	$\frac{1}{(s^2+\omega^2)^2}$